

Chapter 1 System Architecture

Figure 1.1 illustrates the end-to-end (E2E) architecture adopted in this thesis. The system consists of several key components, including a CU/DU node, a core network, a commercial O-RU, and a 5G User Equipment (UE). These components collaboratively establish the 5G communication chain from the core to the radio interface.

To support configuration automation and test result visualization, a custom integration rApp [?] is deployed on the SMO. Test cases are fed into the system through a RESTful API as inputs. The rApp interprets these test cases, dynamically triggers DU/RU configuration via the configuration service, and initiates end-to-end testing. Once the UE successfully attaches and traffic is generated, throughput results are collected and published to Grafana dashboards [?] as output. This automation pipeline enables streamlined and repeatable E2E integration testing.

In this work, the core network implementations tested include free5GC [?] and Open5GS [?]. However, the architecture is designed to support any core network solution capable of interfacing with mobile devices via standard NG signaling.

On the CU/DU side, the system is compatible with any implementation that utilizes a configuration file-driven setup. The OAI gNB has been used as the primary test platform, but the system is not limited to it.

For the RU, it is required to support M-plane management interfaces to allow remote configuration and monitoring. In this study, the JURA RU [?] has been used for validation.

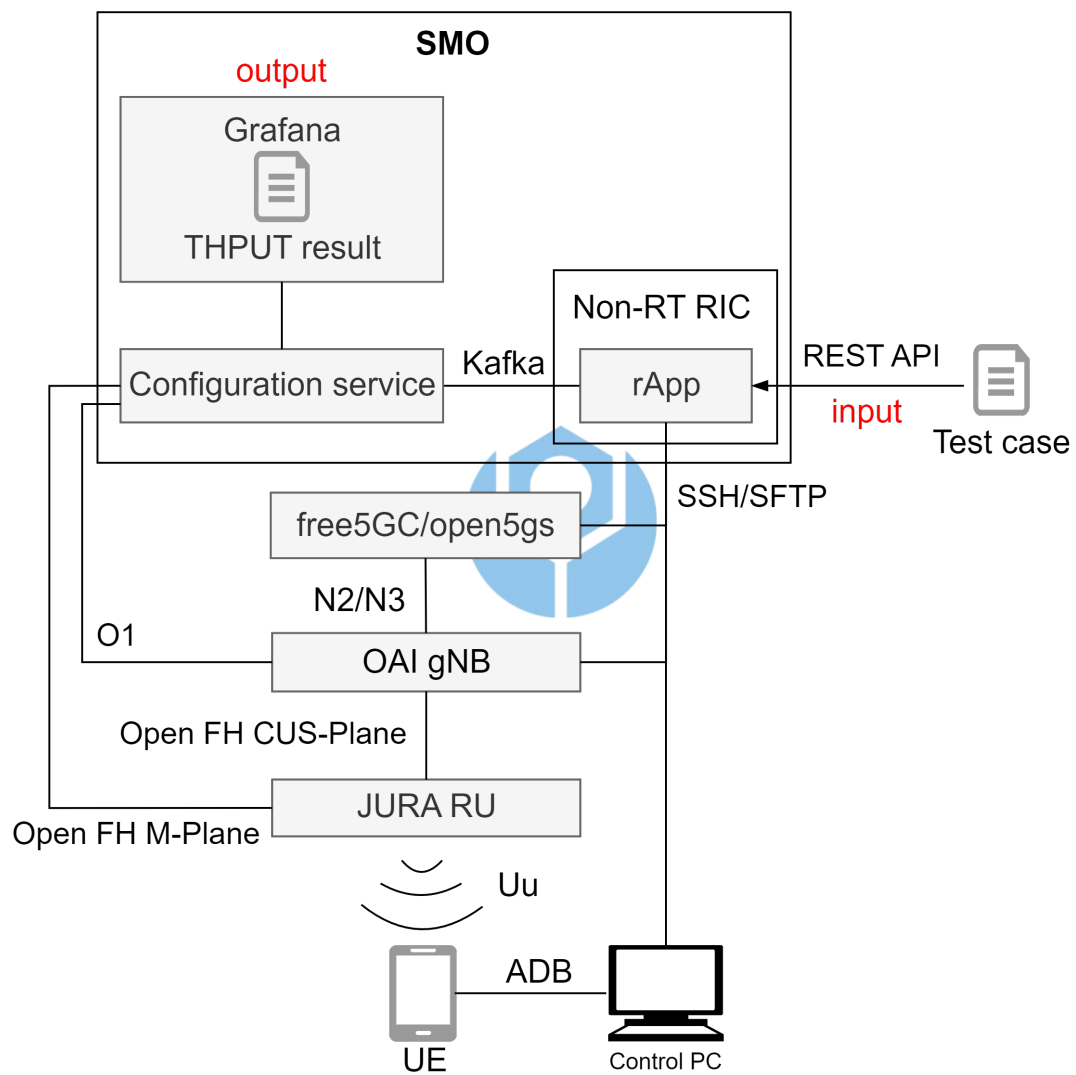


Figure 1.1: System Architecture

Regarding the UE, devices must support control via Android Debug Bridge (adb) commands [?], enabling automated access procedures and performance testing. Here we use MTK UE [?].

